

5 Pronghorns are similar to antelopes and are native to North America.

They live in open areas of grasslands.

The photograph shows a pronghorn.



(Source: © Paul Tessier/Shutterstock)

(a) These pronghorns can run long distances at speeds of 90 km per hour.

(i) Which of the following features of pronghorns is an adaptation for running long distances at high speeds?

(1)

- ☐ **A** decreased glomerular filtration rate
- ☐ **B** increased number of slow twitch muscle fibres
- ☐ **C** narrower airways
- ☐ **D** smaller cardiac output

(ii) Explain how the heart rate and ventilation rate of a pronghorn are increased to enable it to run long distances at high speeds.

(4)

- (b) Scientists measured the mass, resting metabolic rate and pulse rate of five mammals.

The table shows the results.

Mammal	Mass / kg	Resting metabolic rate / joules second ⁻¹	Pulse rate / beats per minute
rat	0.260	1.45	420
dog	16.0	20.0	100
sheep	45.0	50.1	70
human	70.0	87.0	72
cow	400	267	40

- (i) Describe the relationships between mass, resting metabolic rate and pulse rate shown in the table.

(3)

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- (ii) When a human exercises, the metabolic rate is increased and heat energy is released.

Describe how the body temperature of a human is maintained at 37°C during exercise.

(3)

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(Total for Question 5 = 11 marks)